Spectral–Spatial Attention-Guided Multi-Resolution Network for Pansharpening

Shen Xu, Shengwei Zhong, *Member, IEEE*, Hui Li, and Chen Gong, *Senior Member, IEEE*

***Abstract*—Pansharpening is a technique that combines high-resolution panchromatic (PAN) images with low-resolution multispectral (MS) images to produce high-resolution MS (HRMS) images. Deep learning-based pansharpening have outperformed traditional methods on detail injection and spectral preserving. However, existing methods often directly learn the mapping between PAN, MS, and fused HRMS, without considering the spectral–spatial feature correlation in separate bands among PAN, low-resolution PAN (LRPAN), and MS. To address this limitation, we propose a novel network called spectral–spatial attention-guided multiresolution network (SSA-MRN). Initially, SSA-MRN incorporates LRPAN images to capture the intermediate features between MS and PAN images. It also uses the individual bands of MS to learn band-specific features. Based on the comprehensive features, the spectral–spatial attention integration (SSAI) module is introduced at various scales. SSAI leverages a dot-product attention mechanism to selectively enhance the associative spectral–spatial features between PAN images and MS images across different spectral bands. The features learned by the SSAI are progressively fused at each resolution to produce the final output. Experiments on two benchmark datasets are conducted at both reduced-resolution and full-resolution. Results demonstrate that our SSA-MRN significantly enhances pansharpening quality compared to five classical methods and four state-of-the-art deep learning-based methods.**

***Index Terms*—Multi-resolution, multi-spectral (MS) feature, pansharpening, spectral–spatial attention integration (SSAI).**

I. Introduction

With the rapid development of technology, the spatial resolution of remote sensing images is increasing. However, due to inherent limitations in imaging, multispectral (MS) images always have a lower spatial resolution compared to panchromatic (PAN) images. Pansharpening combines the rich spectral information from MS images and the high-frequency spatial details from PAN images, resulting in high-resolution MS (HRMS) images. The fused images are widely utilized in applications such as land anomaly detection [1], sensing data applications [2], and crop monitoring [3].

Current pansharpening methods can be broadly categorized into classical methods and deep learning-based methods. Classical pansharpening methods can be further divided into three categories: Component substitution (CS) methods multiresolution analysis (MRA) methods, and variational optimization (VO) methods. CS methods separate the spatial and spectral information of MS image through specific transformations, and then replace the spatial component with PAN image. Common CS methods include IHS fusion [4], principal component analysis [5], Brovey transform [6], and Gram–Schmidt (GS) [7]. Besides, improved CS methods, such as nonlinear IHS and GS adaptive (GSA) can reduce spectral distortion. Although CS methods are fast, they can generate artifacts [7]. MRA methods extract high-frequency details from the PAN image through multiresolution decomposition and injects them into the upsampled MS image. Typical MRA methods include discrete Wavelet transform [8], high-pass filtering fusion [9], indusion method [10], Laplacian pyramid (LP) [11], and additive wavelet transform [12]. MRA methods introduce less spectral distortion but may cause local spatial artifacts [11]. VO [13], [38], [39] methods achieve spectral and spatial quality balance by optimizing constraints between HRMS and PAN images. These constraints are typically formulated through two terms: a fidelity term describing the relationship between HRMS and observed images, and a regularization term incorporating prior knowledge about HRMS images into the optimization process. For example, Bayesian probability models [36] apply probabilistic priors to estimate HRMS images from PAN inputs, sparse representation methods [37] reconstruct HRMS images by enforcing sparsity constraints in a dictionary learning framework, and Variational models [35] use energy function minimization techniques to align spatial and spectral features.

In recent years, deep learning-based methods have made significant progress in the field of pansharpening [15], [16], [17], [18], [19], [20]. Generally, these methods involve training a neural network on a large dataset of PAN and MS images to learn a mapping that fuses the two types of images effectively. Yang et al. [14] proposed PanNet, a pansharpening model based on deep convolutional neural networks. PanNet enhances spatial structure fidelity by training in the high-frequency domain

Received 6 November 2024; revised 6 January 2025 and 17 February 2025; accepted 17 February 2025. Date of publication 25 February 2025; date of current version 20 March 2025. This work was supported in part by the National Science Foundation (NSF) of China under Grant 62101261, in part by the NSF of Jiangsu Province under Grant BK20210332, and in part by the Fundamental Research Funds for the Central Universities under Grant 30923011027. *(Corresponding authors: Shengwei Zhong; Hui Li.)*

Shen Xu, Shengwei Zhong, and Chen Gong are with the PCA Laboratory, the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, the Jiangsu Key Laboratory of Image and Video Understanding for Social Security, and the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: shenxu@njust.edu.cn; zhongsw_91@foxmail.com; chen.gong@njust.edu.cn).

Hui Li is with Zhengzhou Tobacco Research Institute of CNTC, Zhengzhou 450001, China (e-mail: lih@ztri.com.cn).

Our code will be available online at https://github.com/zhouchuanxu/SSA-MRN.git after being accepted.

Digital Object Identifier 10.1109/JSTARS.2025.3543827

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

and effectively fuses spectral and spatial information through upsampled MS images. Building on the foundation of convolutional networks, Zhang et al. [22] introduced a multiscale network structure. They proposed a bidirectional multiscale learning pansharpening method, referred to as bidirectional pyramid network. This method designs a pyramid-structured bidirectional network architecture to separately process LRMS images and high-resolution PAN images. Not satisfied with only altering the image scale, researchers further explored features across different spectral bands. Zhou et al. [23] developed PAN-guided band-aware multispectral feature enhancement network (PGFENet), which selectively enhances MS image features and extracts multiscale features using a PAN-guided band-aware enhancement module and a multifocus feature fusion module. In addition, some networks do not directly learn the mapping but instead learn their conditional probability distribution. PanFlowNet [24], a flow-based deep network for pansharpening, employs an invertible neural network with conditional affine coupling blocks to generate diverse HRMS images, using proposed the maximum probability criterion to select the most suitable result.

Attention mechanisms have been increasingly adopted in the field of pansharpening to improve feature extraction and fusion capabilities. An efficient nonlocal attention ResNet for pansharpening, proposed by Zhang et al. [42], enhances computational efficiency by employing 1-D convolution in a nonlocal attention module, which reduces the cost of cross-channel interactions. Other approaches, such as parallel multiscale attention constraint network for pansharpening, proposed by Liang et al. [44], utilize dilated convolutions at various scales and employ learnable layer normalization to adaptively adjust channel weights, enhancing the multiscale feature fusion. PGFENet [23] introduces a multifocus feature fusion attention module that combines parallel and cascaded dilated convolutions to achieve multiscale feature extraction while reducing the parameter load. However, despite the effectiveness of these methods, their attention mechanisms primarily focus on a single input, either enhancing spatial or spectral features independently, or performing multiscale feature extraction. This direct approach often overlooks the valuable features present in the low-resolution PAN (LRPAN) and fails to exploit the spectral–spatial feature correlation between PAN and individual bands of MS. This oversight can significantly compromise the overall performance of the network, leading to suboptimal image fusion results.

To address this limitation, we propose a spectral–spatial attention-guided multiresolution network (SSA-MRN), which extracts spectral–spatial features from MS and PAN images at different scales and progressively integrates them into the final HRMS output. We propose incorporating LRPAN images into the network to produce rich intermediate features between MS and PAN images. These features complement the high-resolution details from PAN and the spectral information from MS images, leading to a more comprehensive feature set. The spectral–spatial attention integration (SSAI) module is the core component of this network. Unlike other attention mechanisms that primarily operate on a single input, the SSAI module is specifically designed for the pansharpening task to leverage

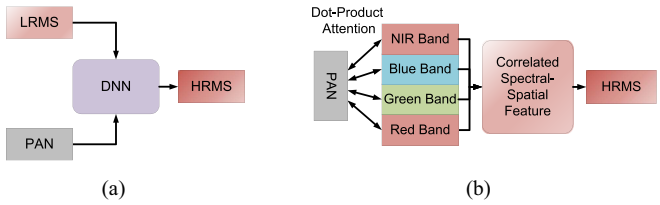


Fig. 1. Comparison between (a) typical deep learning-based methods and (b) proposed SSA-MRN.

both the LRMS images and the PAN images. This module employs a dot-product mechanism to effectively capture the spectral–spatial correlations between the PAN and MS images. We divide the MS images into multiple spectral bands and extract the features of each band individually. By learning the relational information between LRMS and PAN, the SSAI module integrates these features to enhance the various characteristics of the pansharpened images. This task-specific design distinguishes the SSAI module from generic plug-and-play attention mechanisms. The comparison between existing methods [14], [25], [26] and our method is shown in Fig. 1. In summary, our main contributions are described as follows.

1) We propose a novel pansharpening network, SSA-MRN. SSA-MRN introduces LRPAN images to bridge the feature extraction process between MS and PAN images. SSA-MRN integrates spectral–spatial features from both MS and PAN images across multiple resolutions, resulting in a more comprehensive and enriched feature set. This integration is performed on a bandwise basis to capture band-unique spectral information from MS.
2) We develop the SSAI module that employs a dot-product attention mechanism to refines the integration of PAN and MS images. SSAI selectively focus on the most relevant features between PAN at different resolutions and the spectral bands of MS. This enables the network to better capture the relationships between the spatial and spectral features and inject this valuable information into the network.
3) We conducted comprehensive experiments, demonstrating that SSA-MRN achieves superior performance on both 4-band and 8-band satellite images. In addition, SSA-MRN exhibits better cross-satellite generalization capability compared to SOTA deep learning methods.

II. Proposed Method

In this section, detailed information on our proposed SSA-MRN method is provided. Our method focuses on addressing the pansharpening problem by fusing high-resolution PAN images with MS images. We detail the proposed method, as illustrated in Fig. 2. The network features a novel overall structure where PAN, LRPAN, and MS images are first incorporated to produce comprehensive features. The network then progressively employs the new SSAI module to produce correlated spectral–spatial features at various resolutions. This module employs a dot-product attention mechanism to selectively inject the correlated features between different spectral bands and PAN images

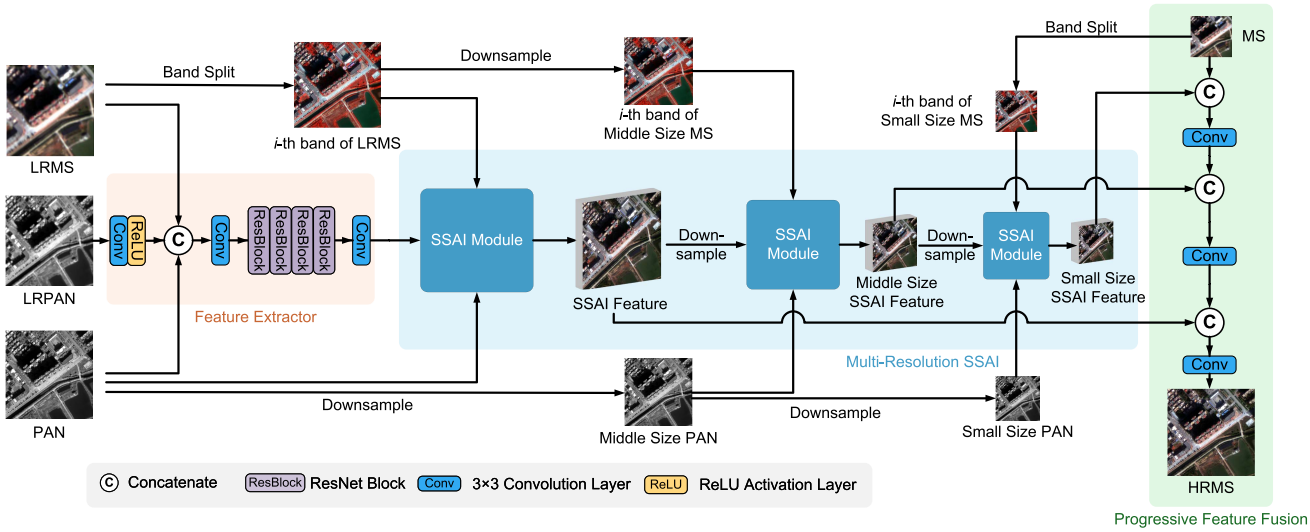


Fig. 2. Overall structure of SSA-MRN. It consists of three integral components: the feature extraction part, multiresolution SSAI, which integrates PAN images of different scales and MS images of different spectral bands, and the progressive feature fusion part, which integrates features of different scales in sequence.

into the network. The overall structure is designed for datasets with the scale ratio of four, and it can be easily expanded to datasets with other resolution ratios.

A. Problem Formulation

The PAN image is represented as $\mathbf{P} \in \mathbb{R}^{H\times W}$, where H and W denote the height and width of the PAN image, respectively. The original MS image is denoted as $\mathbf{M}_s \in \mathbb{R}^{h\times w\times C}$, with h and w denoting the height and width of the original MS image, and C representing the number of spectral bands. Typically, the ratio between H and h is 4. To match the dimensions of the PAN image, we upsample the MS image using bicubic interpolation, resulting in what we refer to as LRMS image $\mathbf{M}_l \in \mathbb{R}^{H\times W\times C}$.

In this work, we chose to use bicubic interpolation [21] for tasks, such as matching dimensions between PAN and MS images. Bicubic interpolation offers a balance between preserving spatial details and providing smoother transitions by using a larger set of surrounding pixels for the weighted average. This method provides better visual quality compared to nearest neighbor interpolation, which can result in abrupt transitions and cause "blockiness." While bilinear interpolation is computationally simpler, it may not preserve image details as effectively, especially when working with small patch sizes, such as the 16×16 patches used in our training data.

B. Overall Network Architecture

The overall network framework consists of three components: the feature extraction part, multiresolution SSAI, and the progressive feature fusion part. In addition to the conventional PAN and LRMS inputs, LRPAN image is introduced as an additional feature input to capture spatial features distinct from those of the original PAN. The backbone network is designed to initially extract overall features from the MS, PAN, and LRPAN image inputs. The LRMS image, with its superior spectral information, is then directly incorporated into the detail learning output of the network. This integration leverages the rich spectral content of the MS image to enhance the reconstructed image quality. The feature extraction part is predominantly composed of four residual network (ResNet) blocks [14]. The ability of these blocks to perform advanced feature extraction is integral to the model. ResNet blocks are chosen for their efficacy in mitigating the vanishing gradient problem, and their capability to preserve the high-frequency spatial details that are essential for detailed image reconstruction. $\mathbf{F}_{\text{in}}$ represents the initial features extracted by the network

$$\mathbf{F}_{\text{in}} = f_w(\mathbf{M}_l, \mathbf{P}, \mathbf{P}_l) \tag{1}$$

where f_w is the overall feature extraction and $\mathbf{P}_l \in \mathbb{R}^{H\times W}$ represents LRPAN image, which by first downsampling the PAN image by a factor of four and then upsampling it by the same factor.

The obtained feature $\mathbf{F}_{\text{in}}$ is then input into the SSAI module. In addition, the different bands of $\mathbf{M}_l$ are also used as inputs to this module. This allows the module to leverage the rich spectral information inherent in the MS bands, further refining the feature extraction process. The details of obtaining $\mathbf{F}_{\text{SSAI}}$ will be described in Section II-C. The formula is expressed as follows:

$$\mathbf{F}_{\text{SSAI}} = f_{\text{SSAI}}(\mathbf{F}_{\text{in}}, \mathbf{M}_l^i, \mathbf{P}) \tag{2}$$

where f_{SSAI} represents the SSAI module, which injects the associative information of the unique features from different bands and different sizes of PAN through a dot-product attention into the network. $\mathbf{M}_l^i$ represents the ith band of the LRMS image. Given that the ratio between MS and PAN images is 4, we downsample $\mathbf{F}_{\text{SSAI}}$ by a factor of 2, creating an intermediate feature map that facilitates the alignment of MS and PAN images in subsequent operations. This downsampled feature map, denoted as $\mathbf{F}_{\text{SSAI}}^{\downarrow}$, is then used as input for the following layers of the network. This step allows the network to learn multiscale information, which is crucial for enhancing its feature extraction

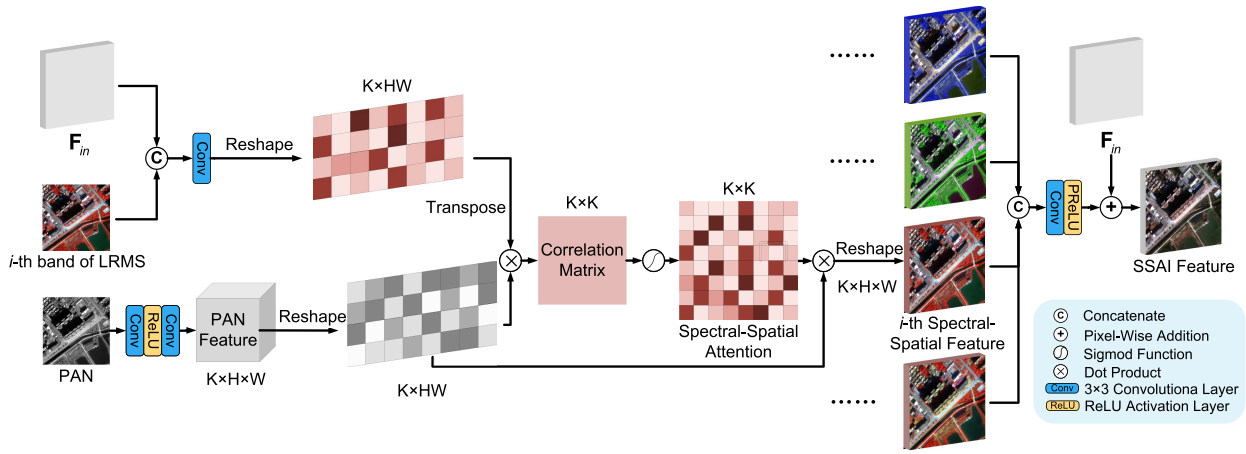


Fig. 3. SSAI module. This module integrates PAN images of different scales and MS images of different spectral bands using a dot-product attention mechanism. The process includes: 1) reshaping spatial and spectral features into 2-D features; 2) computing the correlation matrix using dot-product attention; 3) applying PAN guidance; 4) concatenating the ith features to obtain the final output.

capabilities. The formula for middle SSAI features $\mathbf{F}_{\text{SSAI_}m}$ is expressed as follows:

$$\mathbf{F}_{\text{SSAI_}m} = f_{\text{SSAI}}(\mathbf{F}^{\downarrow}_{\text{SSAI}}, \mathbf{M}^{i}_{m}, \mathbf{P}_{m}) \tag{3}$$

in which $\mathbf{M}^{i}_{m} \in \mathbb{R}^{2h\times 2w\times C}$ represents the ith band of original MS image upsampled by a factor of two, and $\mathbf{P}_{m} \in \mathbb{R}^{2h\times 2w}$ represents the PAN image downsampled by a factor of two. The choice of a factor of two for upsampling and downsampling based on the fact that generally, a bundle of MS and PAN images are captured with a ratio of four. By using a factor of two, we can upsample the MS image once more to align it with the PAN image, allowing for easier and more efficient matrix operations. We further downsample $\mathbf{F}_{\text{SSAI_}m}$ by a factor of two to align with the original MS image dimensions, which results in the small feature map denoted as $\mathbf{F}^{\downarrow}_{\text{SSAI_}m}$. Subsequently, we pass this feature map through the final SSAI module to obtain the small SSAI feature $\mathbf{F}_{\text{SSAI_}s}$

$$\mathbf{F}_{\text{SSAI_}s} = f_{\text{SSAI}}(\mathbf{F}^{\downarrow}_{\text{SSAI_}m}, \mathbf{M}^{i}_{s}, \mathbf{P}_{s}) \tag{4}$$

where $\mathbf{P}_{s}$ is obtained by downsampling $\mathbf{P}$ by a factor of four.

Subsequently, features are progressively fused at various scales. The small feature $\mathbf{F}_{s}$ is obtained as follows:

$$\mathbf{F}_{s} = \text{Conv}_{3\times 3}(\text{Concat}(\mathbf{M}_{s}, \mathbf{F}_{\text{SSAI_}s})) \tag{5}$$

where $\text{Conv}_{3\times 3}(\cdot)$ represents a 3×3 convolutional layer, and $\text{Concat}(\cdot,\cdot)$ denotes the concatenation of the channels of the two inputs. After this concatenation and convolution operation, the resulting feature map captures both the fine spatial details from the original MS image $\mathbf{M}_{s}$ and the small SSAI features $\mathbf{F}_{\text{SSAI_}s}$. We upsample $\mathbf{F}_{s}$ by a factor of two, denoted as $\mathbf{F}^{\uparrow}_{s}$, the midresolution fusion output $\mathbf{F}_{m}$ is obtained as follows:

$$\mathbf{F}_{m} = \text{Conv}_{3\times 3}(\text{Concat}(\mathbf{F}^{\uparrow}_{s}, \mathbf{F}_{\text{SSAI_}m})). \tag{6}$$

We upsample $\mathbf{F}_{m}$ by a factor of two, denoted as $\mathbf{F}^{\uparrow}_{m}$, which is then used as input for the next progressive stage. The final output $\mathbf{HRMS}$ is obtained by another convolutional operation on the concatenated features of $\mathbf{F}^{\uparrow}_{m}$ and $\mathbf{F}_{\text{SSAI}}$

$$\mathbf{HRMS} = \text{Conv}_{3\times 3}(\text{Concat}(\mathbf{F}^{\uparrow}_{m}, \mathbf{F}_{\text{SSAI}})). \tag{7}$$

By leveraging upsampling, concatenation, and convolution operations, the network effectively integrates multiscale spatial and spectral features to produce HRMS images.

C. SSAI Module

In traditional approaches, features from PAN images and MS images are either extracted independently or directly concatenated along the channel dimension without adequately accounting for their interrelated characteristics. Given the inherent correlation between PAN and MS features, it is essential to employ a more sophisticated method to fuse the information from both sources. To address this, we propose using dot-product attention mechanism to extract associative information from PAN and MS images, assigning appropriate spectral–spatial weights based on feature relevance for better data fusion. In addition, by separating different spectral bands, we explore and extract the unique relationships between each band and the PAN image. We detail the proposed module as illustrated in Fig. 3.

First, $\mathbf{F}_{\text{in}}$ obtained from the feature extraction part and $\mathbf{M}^{i}_{l}$ along the channel dimension are concatenated as the input feature cube for our module. The spectral feature $\mathbf{F}_{i}$ of the SSAI module is expressed as follows:

$$\mathbf{F}_{i} = \text{Conv}_{3\times 3}(\text{Concat}(\mathbf{F}_{\text{in}}, \mathbf{M}^{i}_{l})) \tag{8}$$

where a 3×3 convolution $\text{Conv}_{3\times 3}(\cdot)$ is used to adjust the number of channels of the concatenated feature. The dot-product attention mechanism specifically reshapes the combined features $\mathbf{F}_{i}$ into a 2-D feature $\mathbf{C}^{i}_{\text{ms}} \in \mathbb{R}^{K\times HW}$. The channel dimension K represents the number of features. In addition, we incorporate PAN to enhance the associative information between the two image types. To achieve this, we extract features from PAN using two 3×3 convolutional layers with a ReLU activation function applied in between. This approach ensures that the

deep features of PAN are adequately captured and utilized to improve the effectiveness of the dot-product attention mechanism. The spatial feature $\mathbf{F}_{\text{pan}}$ of the SSAI module is expressed as follows:

$$\mathbf{F}_{\text{pan}} = \text{Conv}_{3\times3}(\text{ReLU}(\text{Conv}_{3\times3}(\mathbf{P}))). \tag{9}$$

The spatial features obtained after convolution are reshaped into a 2-D feature matrix $\mathbf{C}_{\text{pan}} \in \mathbb{R}^{K\times HW}$ to align with spectral features.

Subsequently, to enhance the associative information between the spectral and spatial features, we perform a matrix multiplication of the obtained the spatial features $\mathbf{C}_{\text{pan}}$ with the transpose of spectral features $\mathbf{C}^i_{\text{ms}}$. The matrix $\mathbf{R}$ is calculated by computing the dot products between each feature vector in $\mathbf{C}^i_{\text{ms}}$ and $\mathbf{C}_{\text{pan}}$. Each row of the obtained matrix $\mathbf{R}$ represents the correlation between a particular channel of $\mathbf{C}_{\text{pan}}$ and all channels of $\mathbf{C}^i_{\text{ms}}$

$$\mathbf{R} = \mathbf{C}_{\text{pan}}\mathbf{C}^{i\ \top}_{\text{ms}}. \tag{10}$$

The formula for obtaining the weight matrix $\mathbf{W}$ can be expressed as follows:

$$\mathbf{W} = \text{Softmax}(\mathbf{R}) \tag{11}$$

where $\text{Softmax}(\cdot)$ represents the softmax function, we then perform an elementwise multiplication of the normalized attention weight matrix $\mathbf{W}$ with $\mathbf{C}_{\text{pan}}$. The spectral–spatial feature $\mathbf{F}^i_{\text{ss}}$ can be expressed as follows:

$$\mathbf{F}^i_{\text{ss}} = \mathbf{W}\mathbf{C}_{\text{pan}} \tag{12}$$

where $\mathbf{F}^i_{\text{ss}}$ represents the ith spectral–spatial feature after dot-product attention mechanism.

Finally, we concatenate the four different features to obtain the overall feature $\mathbf{F}_{\text{combine}}$. To refine the $\mathbf{F}_{\text{combine}}$, a 3×3 convolution is applied to adjust the channel dimensions, followed by a parametric ReLU (PReLU) activation function to introduce nonlinearity

$$\mathbf{F}'_{\text{combine}} = \text{PReLU}(\text{Conv}_{3\times3}(\mathbf{F}_{\text{combine}})). \tag{13}$$

PReLU [27] allows for a more comprehensive retention of features, compared to ReLU, which discards negative values. Given the depth of the network, we directly add the input features $\mathbf{F}_{\text{in}}$ to the refined combined features $\mathbf{F}'_{\text{combine}}$ to prevent gradient explosion and ensure stable training. The final output of the module, denoted as $\mathbf{F}_{\text{SSAI}}$, is given by

$$\mathbf{F}_{\text{SSAI}} = \mathbf{F}_{\text{in}} + \mathbf{F}'_{\text{combine}}. \tag{14}$$

The implementation of the SSAI module is summarized in Algorithm 1.

D. Loss Function

We adopt the mean squared error (MSE) (L2 loss) to optimize our proposed method

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} \|\mathbf{HRMS} - \mathbf{GT}\|_2 \tag{15}$$

where $\mathbf{GT}$ represents ground truth.

Algorithm 1: The Proposed SSAI Module.

Input: Feature extractor result $\mathbf{F}_{in}$; the i-th band of the LRMS image $\mathbf{M}^i_l$; panchromatic image $\mathbf{P}$
Output: Spectral-Spatial Attention Integrated Feature $\mathbf{F}_{SSAI}$

1: **for** each band i **do**
2: Construct the spectral feature $\mathbf{F}_i$ via (6) and spatial feature $\mathbf{F}_{pan}$ via (7);
3: Reshape $\mathbf{F}_i$ into two-dimensional feature $\mathbf{C}^i_{ms}$;
4: Reshape $\mathbf{F}_{pan}$ into two-dimensional feature $\mathbf{C}_{pan}$;
5: Decide the value of $\mathbf{R}$ via (8);
6: Apply softmax to $\mathbf{R}$ to obtain the weight matrix $\mathbf{W}$;
7: Update i-th spectral-spatial attention feature via (10);
8: **end for**
9: Concatenate features $\mathbf{F}^i_{ss}$ to obtain $\mathbf{F}_{combine}$;
10: Get the final output via (12).

III. Experiment

In this section, we conduct several experiments to verify the effectiveness of our proposed SSA-MRN method.

A. Dataset and Experimental Setup

We conducted several experiments using the PanCollection dataset [28], [29], which includes data collected from the 4-band GaoFen-2 (GF2), QuickBird (QB) satellites, and 8-band WorldView-2 (WV2), WorldView-3 (WV3) satellites. The spatial resolutions of the MS and PAN images from GF2 are 4 and 1 m, respectively, while those from QB are 2.44 and 0.61 m, respectively. For 8-band WV3, the dataset consists of MS images with a resolution of 1.24 m and PAN images with a resolution of 0.31 m, whereas WV2 provides 8-band MS images with a resolution of 2 m and PAN images with a resolution of 0.5 m. The dataset segments the original satellite images from GF2, QB, and WV3 into 64×64 MS image patches and 256×256 PAN image patches and further downsamples the MS images to 16×16 patches using the Wald protocol [29]. The QB training set contains 19 044 pairs of images, while the GF2 training set includes 22 010 pairs of images. The WV3 training set consists of 10 794 pairs of eight-channel images. WV2 does not have a training set. All test sets are divided into two settings: reduced-resolution and full-resolution. In the reduced-resolution setting, the PAN image size is 256×256 and the MS image size is 64×64, while in the full-resolution setting, the PAN image size is 512×512 and the MS image size is 128×128. In addition, we evaluated our method using the most commonly used metrics in the pansharpening field, including spectral angle mapper (SAM) [35], erreur relative globale adimensionnelle de synthese (ERGAS) [41], peak signal-to-noise ratio (PSNR), spatial correlation coefficient (SCC), and universal image quality index for 2^n-band images ($Q2^n$) [40] for the downsampled dataset. The SAM metric measures the spectral similarity between the reference and the reconstructed image by calculating the angle between spectral vectors. ERGAS quantifies the relative global

TABLE I
QUANTITATIVE COMPARISON ON THE REDUCED-RESOLUTION AND FULL-RESOLUTION DATASETS (QB DATASET)[1]

Method	SAM ↓	ERGAS ↓	PSNR ↑	SCC ↑	Q4↑	D_λ ↓	D_s ↓	QNR ↑
EXP [34]	8.4593	11.9219	27.7347	0.7164	0.5666	0.0436	0.1502	0.8127
BT-H [6]	7.2112	7.4461	31.7073	0.9148	0.8240	0.2302	0.1648	0.6434
BDSD-PC [30]	8.1025	7.5694	31.5700	0.9050	0.8304	0.1975	0.1636	0.6722
C-GSA [7]	7.2612	7.4305	31.7179	0.9117	0.8310	0.2039	0.2042	0.6341
MTF-GLP-FS [32]	7.8010	7.4119	31.7164	0.9019	0.8207	0.0465	0.1500	0.8108
TV [33]	9.0779	8.6611	30.7108	0.8811	0.8527	0.0550	0.1009	0.8501
PanNet [14]	5.4491	5.3546	35.3892	0.9493	0.8961	0.0438	0.0376	0.9209
FusionNet [25]	5.2740	4.6968	36.4826	0.9602	0.9062	0.0393	0.0397	0.9234
PanFLowNet [24]	5.1391	4.2526	37.2574	0.9678	0.9214	0.0424	**0.0294**	0.9300
PGFENet [23]	4.9243	4.1227	37.5408	0.9696	0.9227	0.0425	0.0434	0.9166
Ours	**4.8478**	**4.0726**	**37.6645**	**0.9702**	**0.9243**	**0.0341**	0.0360	**0.9311**

error by considering both spatial and spectral differences. PSNR evaluates the quality of the reconstructed image based on the ratio between the maximum possible signal and the noise. SCC assesses the spatial correlation between the reference and reconstructed images. For the full-resolution dataset, we used the quality with no reference (QNR) metric, as well as spatial distortion (D_s) and spectral distortion (D_λ) indices,b23. For comparison, we tested several methods, including traditional machine learning methods as follows.

1) **EXP**: MS image interpolation using a polynomial kernel with 23 coefficients [34].
2) **Traditionnal methods.**
   - BT-H: Optimized Brovey transform with haze correction [6].
   - BDSP-PC: Band-dependent spatial-detail with physical constraints [30].
   - C-GSA: Context-based GSA with local parameter estimation exploiting clustering [7].
   - MTF-GLP-FS: Generalized LP with MTF-matched filters with a full scale regression-based injection model [32].
   - TV: Pansharpening based on total variation [33].
3) **DL-based methods**
   - PanNet: ResNet with PAN and LRMS concatenated as input. Each of the four ResNet blocks contains 64 channels [14].
   - FusionNet: ResNet with PAN-LRMS difference as input. Each of the four ResNet blocks contains 32 channels [25].
   - PanFlowNet: Reversible neural networks with conditional affine coupling blocks, where the most suitable result is selected using the proposed maximum probability criterion from the generated diverse HRMS images [24].
   - PGFENet: U-Net-based architecture with spectral guidance and multiscale feature modules [23].

The first five traditional methods were implemented using the public MATLAB toolbox [29]. All deep learning-based models were trained on a GeForce RTX 2080Ti GPU. We trained the models using the loss function described in their respective papers. In addition, we used the Adam optimizer with a consistent batch size of 32, a learning rate of 1×10^{-4}, and trained for 100 epochs. In the SSAI module, the hyperparameter K is fixed at 6, as determined through experimental analysis in Section III-D.

B. Quantitative Assessments

Table I presents a comprehensive comparison of the quality evaluations between SSA-MRN and existing techniques. The experiments were conducted on the QB and GF2 datasets, encompassing both reduced-resolution and full-resolution test sets. The best performance results are highlighted in bold, while the second-best results are underlined for clarity. As illustrated in the table, deep learning-based methods generally outperform traditional approaches on the test sets. Notably, our network SSA-MRN achieves superior performance on the GF2 test set in most cases. This highlights the efficacy of the method in preserving spatial details while minimizing spectral distortion. Specifically in Table II, the EXP method [34] shows the best D_λ performance. We believe that, compared to the QB satellite, the MS images provided by the GF2 satellite inherently have lower resolution. This lower MS resolution may lead to greater smoothing of details in the GF2 MS images at larger scales. Such a smoothing effect can sometimes reduce high-frequency noise, resulting in better spectral fidelity. In other words, larger pixels may average the spectral information more effectively, reducing spectral noise in individual pixels. Furthermore, the EXP method simply upscales the LRMS image without injecting additional spatial details. As a result, it retains most of the original spectral characteristics, leading to superior D_λ performance.

To ensure the comprehensiveness of the experiments, we conducted additional tests on 8-band satellite data from WV3. As shown in Table III, SSA-MRN achieves the best performance in terms of SAM and D_s metrics, demonstrating the effectiveness of the proposed bandwise SSAI module in preserving spectral fidelity. To analyze the cross-satellite generalization ability of the model, all deep learning-based methods were trained on WV3 but tested on WV2. The results, presented in Table IV, reveal a significant drop in quantitative metrics for PanNet [14]

[1]Best results are marked with bold font, and second-best results are marked with underline.

TABLE II
QUANTITATIVE COMPARISON ON THE REDUCED-RESOLUTION AND FULL-RESOLUTION DATASETS (GF2 DATASET)[1]

Method	SAM ↓	ERGAS ↓	PSNR ↑	SCC ↑	Q4 ↑	D_λ ↓	D_s ↓	QNR ↑
EXP [34]	1.8233	2.3693	31.0319	0.8643	0.7999	0.0180	0.0927	0.8881
BT-H [6]	1.6531	1.5297	34.9191	0.9549	0.9176	0.0602	0.1313	0.8165
BDSD-PC [30]	1.6850	1.6573	34.1009	0.9503	0.9013	0.0749	0.1535	0.7833
C-GSA [7]	1.6636	1.6074	34.5303	0.9505	0.9101	0.0697	0.1474	0.7935
MTF-GLP-FS [32]	1.6616	1.5966	34.3433	0.9439	0.9071	0.0346	0.1429	0.8276
TV [33]	1.8719	2.0241	32.9192	0.9427	0.8670	0.1135	0.1358	0.7661
PanNet [14]	1.1316	0.9306	45.9679	0.9892	0.9681	0.0418	0.0563	0.9043
FusionNet [25]	1.3663	1.7965	40.3076	0.9622	0.8847	0.0538	**0.0445**	0.9043
PanFLowNet [24]	1.5792	1.3004	43.0577	0.9773	0.9340	0.0473	0.0682	0.8880
PGFENet [23]	0.9652	0.8943	46.5164	0.9895	0.9629	0.0396	0.0518	0.9107
Ours	**0.9434**	**0.8755**	**46.9734**	**0.9898**	**0.9688**	**0.0395**	0.0487	**0.9137**

TABLE III
QUANTITATIVE COMPARISON ON THE REDUCED-RESOLUTION AND FULL-RESOLUTION DATASETS (WV3 DATASET)[1]

Method	SAM ↓	ERGAS ↓	PSNR ↑	SCC ↑	Q4 ↑	D_λ ↓	D_s ↓	QNR ↑
EXP [34]	5.7534	7.1302	26.8332	0.7434	0.6011	0.0647	0.2470	0.7043
BT-H [6]	4.8734	4.5503	30.9528	0.9253	0.8199	0.1214	0.1493	0.7568
BDSD-PC [30]	5.4024	4.6733	30.6642	0.9075	0.8196	0.1006	0.1370	0.7870
C-GSA [7]	5.6270	4.8463	30.2036	0.8961	0.8064	0.0793	0.1358	0.7956
MTF-GLP-FS [32]	5.2791	4.6775	30.6126	0.9007	0.8199	0.1031	0.1390	0.7833
TV [33]	5.6470	4.7438	30.1907	0.8979	0.8155	0.0943	0.1323	0.7992
PanNet [14]	3.7819	2.7511	37.0201	0.9725	0.8873	0.0314	0.0683	0.9029
FusionNet [25]	3.7693	2.8383	36.7393	0.9691	0.8817	0.0321	0.0814	0.8899
PanFLowNet [24]	4.0378	2.9007	36.4894	0.9618	0.8598	**0.0275**	0.0770	0.8979
PGFENet [23]	3.6362	2.7184	37.2198	0.9729	0.8892	0.0287	0.0708	0.9030
Ours	**3.4873**	**2.5866**	**37.5691**	**0.9735**	**0.8930**	0.0329	**0.0617**	**0.9077**

TABLE IV
QUANTITATIVE COMPARISON ON THE REDUCED-RESOLUTION AND FULL-RESOLUTION DATASETS (TRAINED ON WV3 DATASET TESTED ON WV2 DATASET)[1]

Method	SAM ↓	ERGAS ↓	PSNR ↑	SCC ↑	Q4 ↑	D_λ ↓	D_s ↓	QNR ↑
PanNet [14]	8.5895	5.4326	27.8065	0.9180	0.8062	**0.0610**	0.0712	0.8722
FusionNet [25]	10.5830	12.4565	20.7745	0.8837	0.5690	0.1639	0.1943	0.6746
PanFLowNet [24]	6.5561	5.1060	28.8385	0.9207	0.7913	0.0662	0.0674	0.8708
PGFENet [23]	5.9327	4.8038	29.2795	0.9185	0.8145	0.0636	0.0613	0.8791
Ours	**5.8845**	**4.7254**	**29.4148**	**0.9217**	**0.8215**	0.0657	**0.0549**	**0.8831**

when evaluated on the reduced-resolution test set. Specifically, the ResBlock layers of FusionNet [25] are limited to only 32 channels, which hampers its ability to effectively process both reduced-resolution and full-resolution data. This low channel count restricts the capacity for feature extraction and information fusion of the network, preventing it from adequately capturing and utilizing the rich detail information present in high-resolution images. Both of networks employ single-resolution structures, which exhibit clear disadvantages when tested on unseen datasets. In contrast, SSA-MRN consistently outperforms the compared deep learning-based methods on unseen datasets, further validating its robust generalization capability.

C. Visual Assessments

To provide a more intuitive comparison of the performance differences among various algorithms, we selected four representative samples from the QB and GF2 datasets, including both reduced-resolution and full-resolution images. These samples were used to compare the pansharpening details produced by different algorithms at varying resolutions. Given that the MS images consist of four bands; we utilized the RGB bands for visualization purposes. The bottom row of each sample displays the MSE between the pansharpened results and the ground truth, providing a clearer visualization of the differences in results obtained through different methods [28].

As depicted in Fig. 4, subtle spectral details, such as the small red car located in the lower left corner of the green box in the images, tend to be distorted in the outputs of methods, such as BT-H [6], PanNet [14], and PanFlowNet [24]. In contrast, our method and PGFENet [23] successfully retain diverse spectral information by effectively processing different bands, highlighting the advantages of bandwise processing. Fig. 5

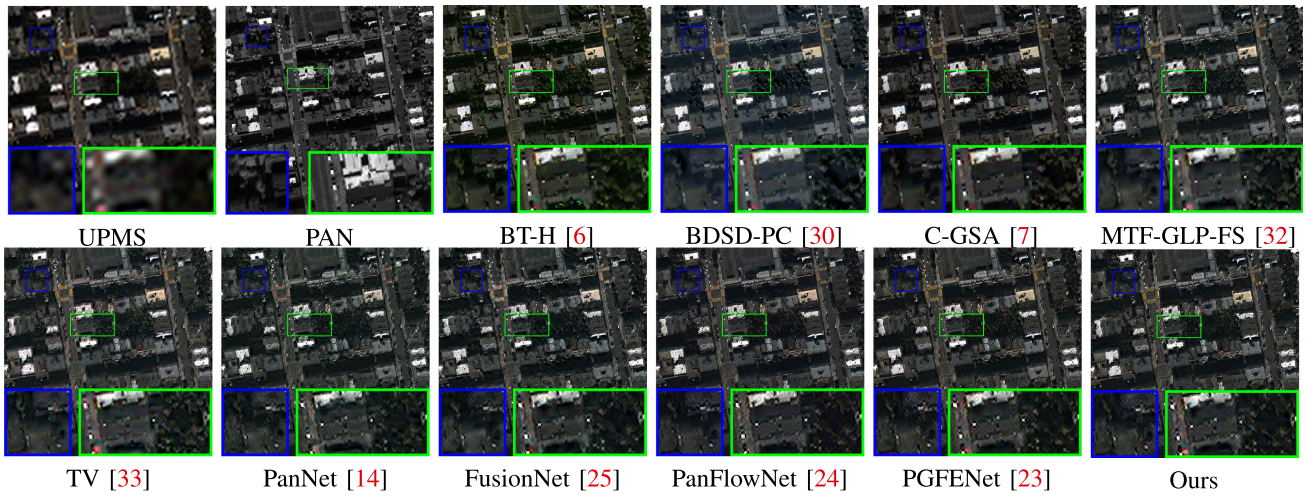


Fig. 4. Full-resolution visualization of images from QB satellite.

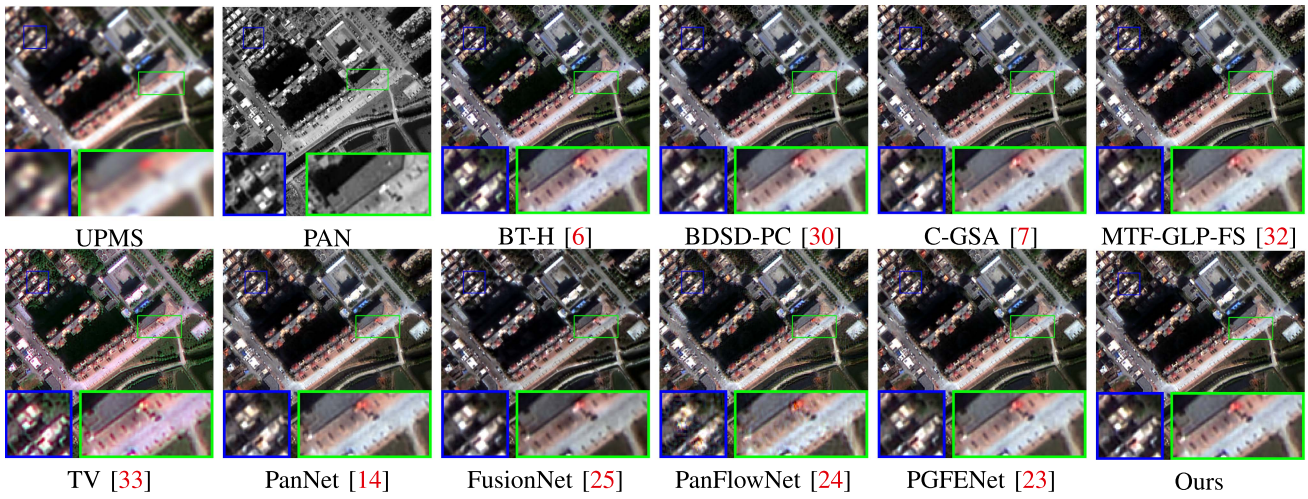


Fig. 5. Full-resolution visualization of images from GF2 satellite.

illustrates the performance of traditional methods, which exhibit significant spectral distortion. After the sharpening of satellite images, large-scale street details often appear blurred. Notably, the TV and PanFlowNet [24] methods show obvious spectral distortions. The deep learning-based method FusionNet [25] tends to produce overly dark results. In contrast, our SSA-MRN demonstrates more stable and high-definition pansharpening results. Among deep learning-based methods, PanFlowNet performs notably better on the QB full-resolution training set compared to the reduced-resolution set. As shown in Fig. 6, unlike the 4-band images, the visualized results of 8-band images produced by traditional methods exhibit significant spectral distortions. This issue is particularly evident in methods, such as C-GSA [7] and TV [33]. In addition, PanNet [14] generates colors that are inconsistent with other images. In contrast, SSA-MRN not only preserves good spectral characteristics but also produces sharper and more well-defined building edges.

Figs. 7 and 8 showcase the visualization results on the reduced-resolution dataset. It is evident that, on the test set with the same size as the training set, our method outperforms other techniques, demonstrating superior spatial and spectral quality. This is particularly pronounced in the GF2 dataset, where the use of reduced-resolution techniques accentuates the sensitivity of spectral information in small-sized images. Similarly, we extended the evaluation to the reduced-resolution 8-band test set of WV3 in Fig. 9. The MSE results demonstrate that SSA-MRN also achieves excellent performance on WV3 data. Overall, our method exhibits a remarkable ability in pansharpening, achieving higher fidelity and clarity.

In summary, the SSA-MRN consistently outperforms existing approaches, both traditional and deep learning-based, across various datasets and resolutions. Its ability to effectively preserve spatial details and minimize spectral distortion makes it a robust and reliable choice for pansharpening applications in remote sensing.

D. Sensitivity Analysis

In the SSAI module, the channel dimension of the spectral and spatial 2-D features is a critical hyperparameter. This parameter guides the attention of the network to the spectral–spatial features. To analyze the sensitivity of the SSAI module to this

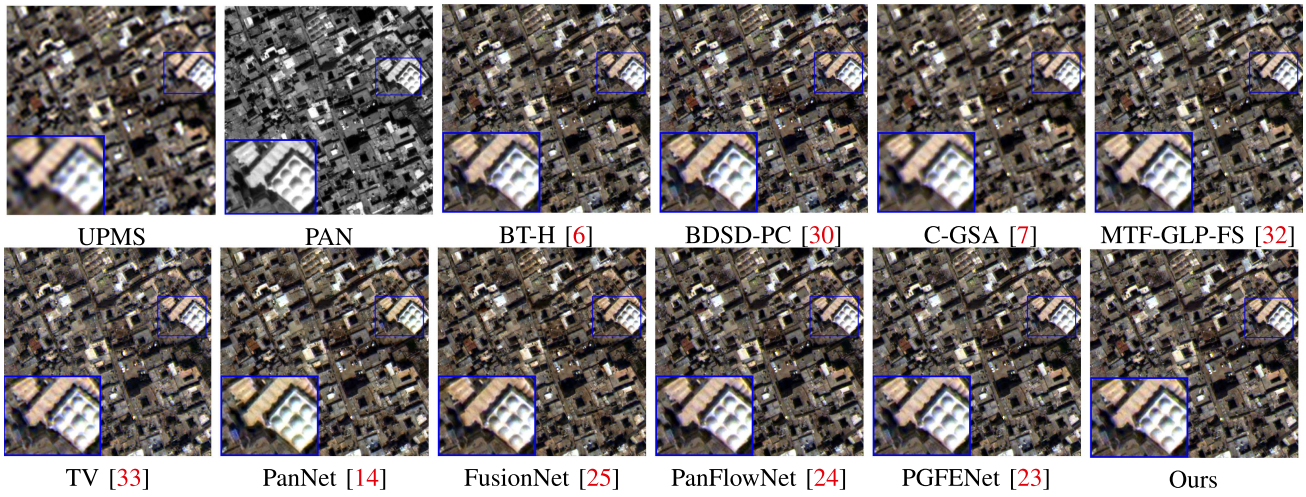


Fig. 6. Full-resolution visualization of images from WV3 satellite.

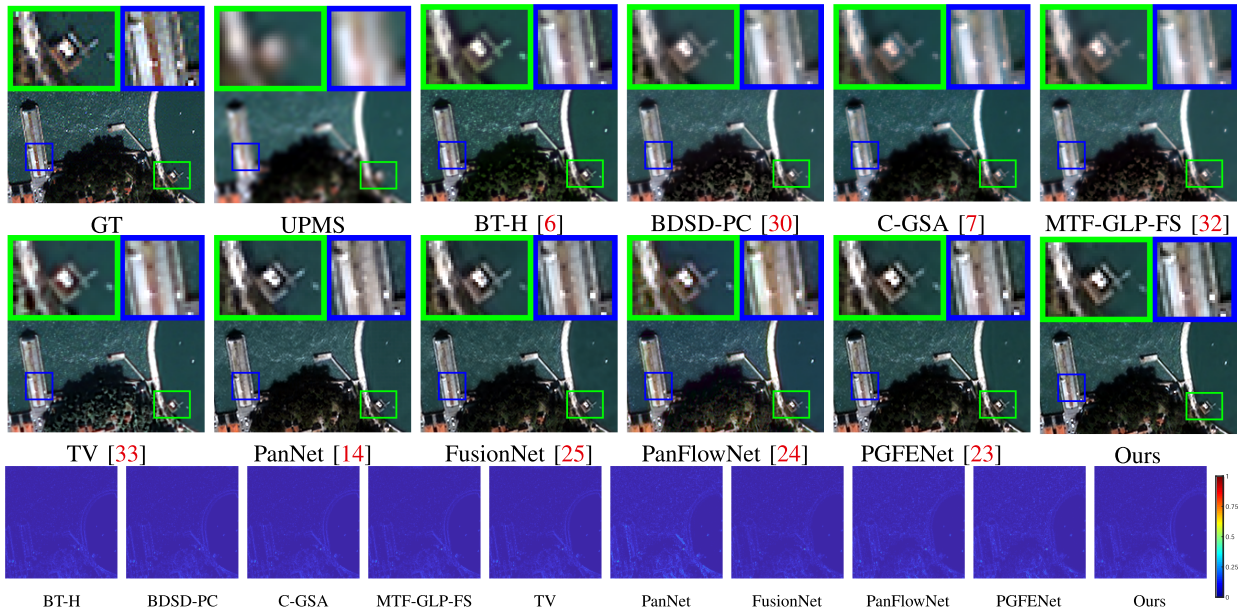


Fig. 7. Reduced-resolution visualization of images from QB satellite and the MSE between the pansharpened results and the ground truth.

channel parameter, we evaluated the quality metrics of the entire network under different channel settings. If the channel dimension is set too low, the integration of spectral–spatial features will be very limited, resulting in minimal performance improvement. Conversely, if it is set too high, the number of feature parameters becomes excessively large, and the network cannot comprehensively integrate the correlated features. Therefore, it is necessary to conduct experiments to select the optimal channel number. For our SSA-MRN, the performance is best when the channel dimension is fixed at 6, as shown in Fig. 10.

E. Ablation Study

To demonstrate the effectiveness of SSA-MRN, we conducted ablation experiments to verify the validity of our overall framework and the SSAI module. Each alternative scheme was tested on the QB satellite dataset.

1) *LRPAN:* To evaluate the role of the LRPAN image in capturing the intermediate features between MS and PAN images, we conducted experiments by removing LRPAN as an input to the network. The results in Table V show a significant drop in performance metrics, indicating that LRPAN serves as an additional feature input to effectively capture spatial features distinct from those of the original PAN image.
2) *Multiresolution Framework (SR):* To evaluate the impact of the multiresolution framework, we replaced the multiresolution structure with a single-resolution framework. Specifically, we removed the upsampling and downsampling operations for various features, such as MS, PAN,

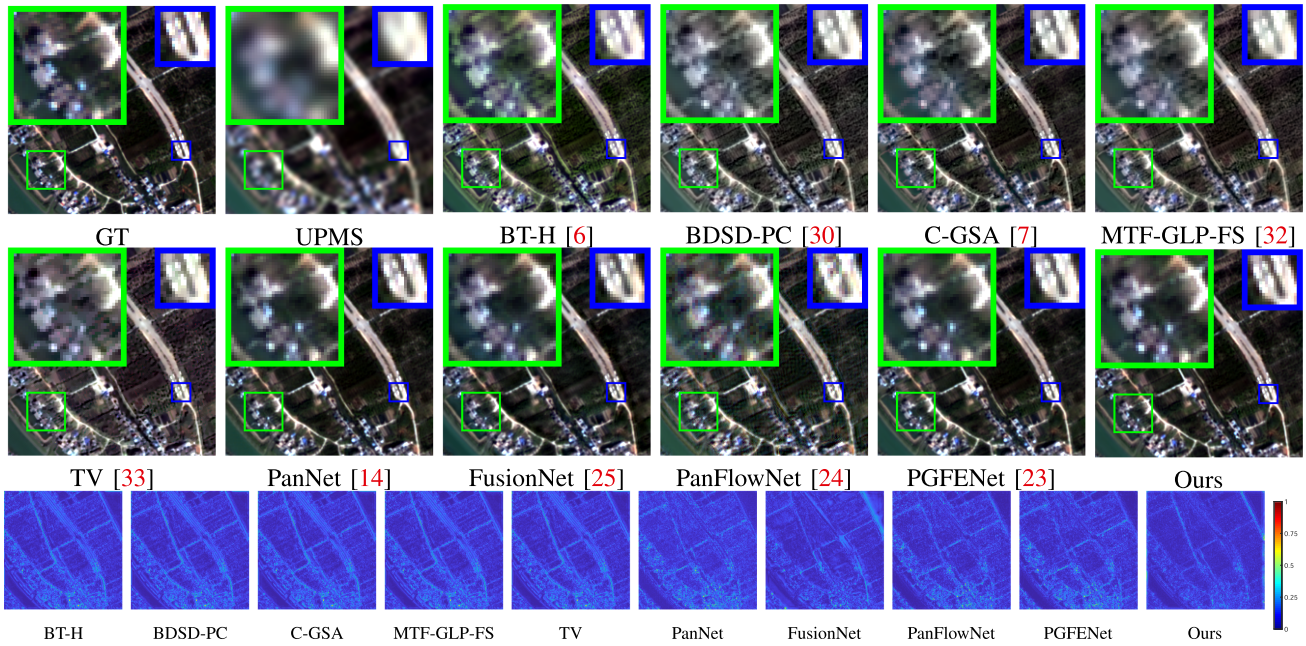


Fig. 8. Reduced-resolution visualization of images from GF2 satellite and the MSE between the pansharpened results and the ground truth.

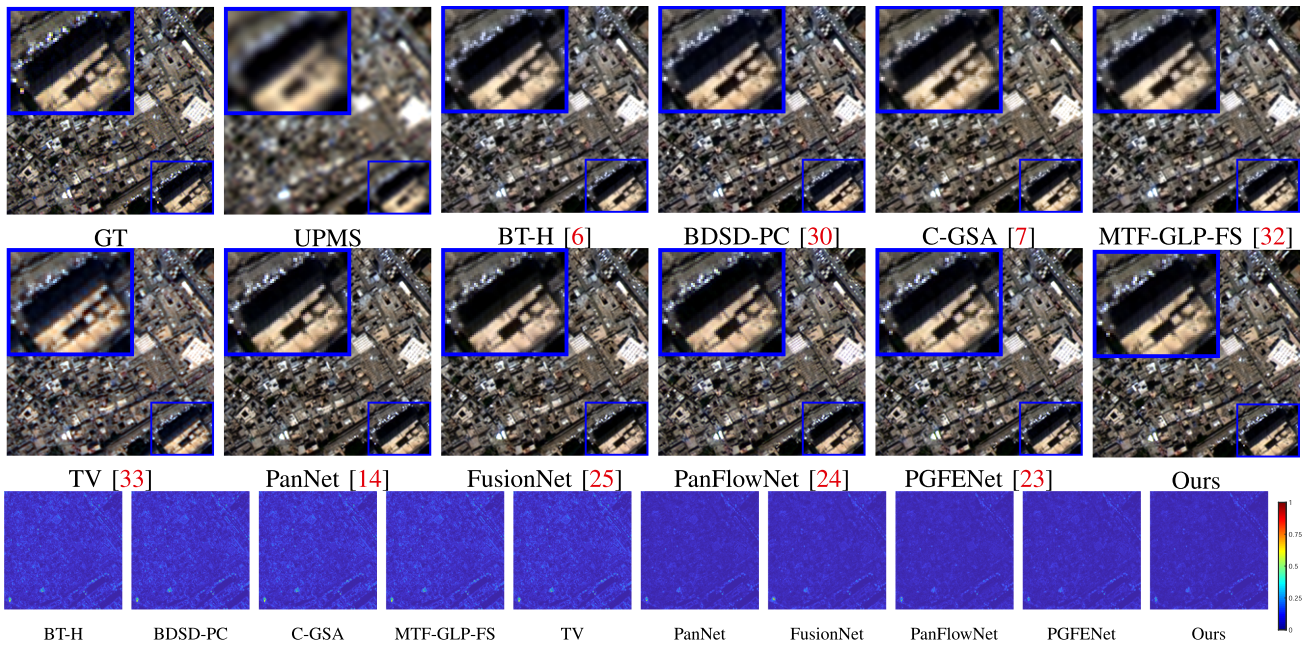
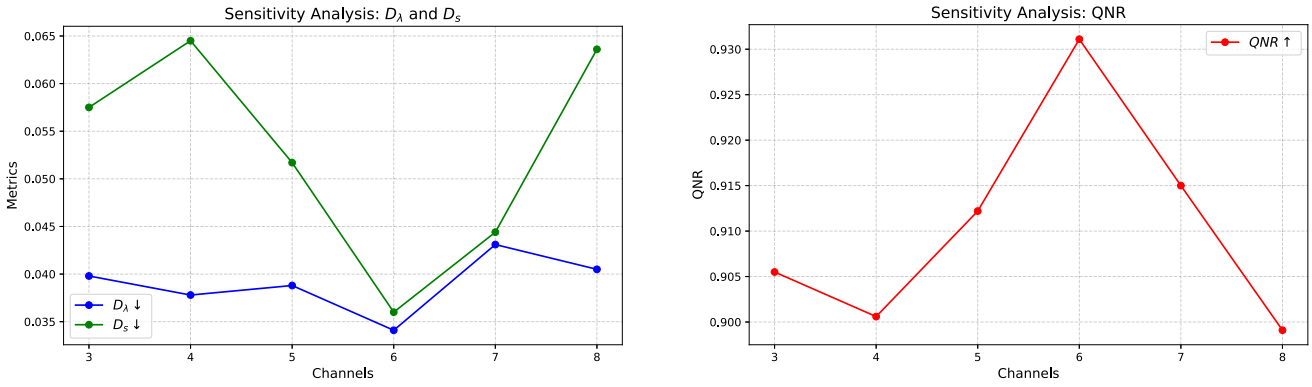


Fig. 9. Reduced-resolution visualization of images from WV3 satellite and the MSE between the pansharpened results and the ground truth.

and SSAI module outputs, within the network. The resolution was kept consistent throughout, while the overall network structure remained unchanged. Compared to D_λ, D_s showed a more significant decline in performance, demonstrating the effectiveness of the multiresolution framework in preserving spatial details on the original resolution test set.

3) *Bandwise Processing (Separate MS):* To assess the effect of bandwise operations, we did not process the MS images by separate bands but instead used the entire MS as the input for the dot-product attention mechanism. The results demonstrate a performance drop, particularly in D_s, indicating that bandwise processing is more effective in preserving spectral characteristics.
4) *SSAI:* To evaluate the impact of correlated features on the overall network, we replaced the dot-product operation with convolutions while keeping the network structure unchanged. Specifically, we avoided reshaping the



(a) (b)

Fig. 10. Sensitivity analysis of SSAI module. (a) D_λ and D_s. (b) QNR.

TABLE V
ABLATION STUDIES[2]

Configuration	No LRPAN	SR	SSAIM	Conv	separate MS	SAM ↓	ERGAS ↓	PSNR ↑	SCC ↑	Q4 ↑	D_λ ↓	D_s ↓	QNR ↑
(I)	✓	×	×3	×	✓	4.9003	4.0830	37.6367	0.9699	0.9231	0.0403	0.0495	0.9128
(II)	×	✓	×3	×	✓	4.8823	4.1743	37.5368	0.9700	0.9213	0.0428	0.0361	0.9232
(III)	×	×	×3	✓	✓	5.2251	7.0456	29.5111	0.8715	0.9146	0.1244	0.0904	0.7963
(IV)	×	×	×3	×	×	4.8964	4.1405	37.1173	**0.9703**	0.9240	0.0417	0.0437	0.9170
(V)	×	×	×2	×	✓	4.8894	4.0865	37.6222	0.9701	0.9241	0.0438	0.0360	0.9225
Ours	×	×	×3	×	✓	**4.8478**	**4.0726**	**37.6645**	0.9702	**0.9243**	**0.0341**	**0.0360**	**0.9311**

Best results are marked with bold font.
No LRPAN: Input without low-resolution PAN.
SR: Implement a single-resolution framework by removing different size.
SSAIM: The number of SSAI modules.
Conv: Remove the dot-product and apply convolutions.
separate MS: Use the entire MS image for the attention instead of processing separate bands.

features into a 2-D form. Instead, the features were directly feeding them into the convolutional layers for learning. The SCC metric, which represents correlation, showed a significant drop, confirming that the proposed SSAI module effectively fuses spectral and spatial correlated features.

5) *SSAI Module Number:* To evaluate the effectiveness of extracting SSAI features at each scale, we removed the middle-scale SSAI features and directly upsampled the fused small-size MS features by four times. The results indicate a significant decrease in the D_λ metric, suggesting that spectral features at the original resolution are affected. This demonstrates the validity of the proposed layer-by-layer fusion approach.

Table V lists the results of the ablation experiments on the QB dataset. The experimental results show that our SSA-MRN performs the best in most metrics, demonstrating the feasibility of our approach. Fig. 11 shows the visual performance of various ablation experiments. The second row of images represents the MSE between the pansharpened results and the ground truth. This visualization clearly highlights the differences between each method and the ground truth, providing a more intuitive comparison of the methods' effectiveness.

[2]Best results are marked with bold font.

F. Efficiency Analysis

To evaluate the computational efficiency of our SSA-MRN, we conducted comprehensive efficiency analysis experiments on both testing and training datasets. For the test set, each method was tested using a set of $20 \times 4 \times 64 \times 64$ MS images and $20 \times 1 \times 256 \times 256$ PAN images as inputs. In addition, we performed efficiency analysis on the QB satellite training dataset, comprising 17 139 pairs of $4 \times 16 \times 16$ MS images and $1 \times 64 \times 64$ PAN images. To ensure a fair comparison of the total training time across different deep learning networks, we trained all models for 100 epochs under the parameter settings specified in the original papers, ensuring that all methods achieved a similar level of convergence. The number of floating-point operations (FLOPs), and the total number of operations, were used as metrics for computational efficiency, as well as the testing time in seconds. The comparative results, as illustrated in Table VI, highlight that FusionNet [25] methods excel due to their simpler network structures. Although our method is not the most optimal in terms of parameter count, the testing time is acceptable and comparable with PanNet [14], PanFlowNet [24], and PGFENet [23]. The parallelizability of our modules ensures high computational efficiency. The high computational efficiency of our method is mainly due to the core SSAI module, which substitutes traditional convolution operations with

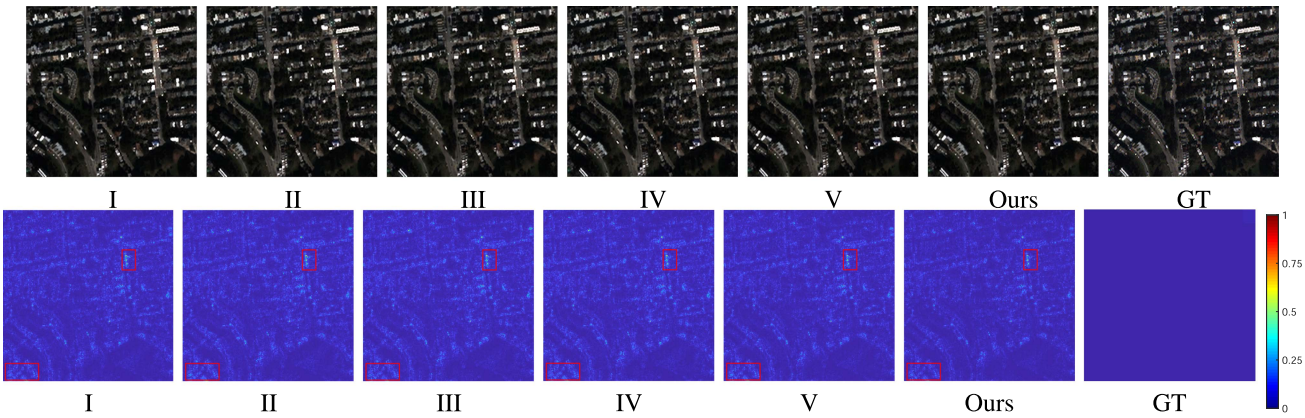


Fig. 11. Pansharpened images resulting from various ablation models and the MSE between the pansharpened results and the ground truth.

TABLE VI
EFFICIENCY COMPARISON. NOP: NUMBER OF PARAMETERS

Methods	Training (min)	Testing (s)	FLOPs (G)	NoP (M)
PanNet	34(GPU)	0.11(GPU)	77.7163	0.5971
FusionNet	13(GPU)	0.04(GPU)	19.4783	0.1503
PanFlowNet	62(GPU)	0.2(GPU)	22.8506	0.0873
PGFENet	45(GPU)	0.15(GPU)	12.7139	0.1525
Ours	43(GPU)	0.14(GPU)	77.8930	0.6037

dot-product attention mechanisms, thereby significantly reducing computational resource requirements.

IV. CONCLUSION

In this article, we propose a novel pansharpening method named SSA-MRN that integrates LRPAN with PAN and MS images to enhance multiresolution feature extraction. In SSA-MRN, SSAI module employs a dot-product attention mechanism to selectively capture spectral–spatial features across different spectral bands. This progressive integration approach significantly improves pansharpening quality, as demonstrated by superior performance on two benchmark datasets compared to both classical and deep learning-based methods.

REFERENCES

[1] T.-J. Zhang, L.-J. Deng, T.-Z. Huang, J. Chanussot, and G. Vivone, "A triple-double convolutional neural network for panchromatic sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9088–9101, Nov. 2023.

[2] J. Cai and B. Huang, "Super-resolution-guided progressive pansharpening based on a deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5206–5220, Jun. 2021.

[3] F. Bovolo, L. Bruzzone, L. Capobianco, A. Garzelli, S. Marchesi, and F. Nencini, "Analysis of the effects of pansharpening in change detection on VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 1, pp. 53–57, Jan. 2010.

[4] W. J. Carper, T. M. Lillesand, and R. W. Kiefer, "The use of intensity-hue-saturation transformations for merging SPOT panchromatic and multispectral image data," *Photogrammetric Eng. Remote Sens.*, vol. 56, no. 4, pp. 459–467, Apr. 2004.

[5] V. P. Shah, N. H. Younan, and R. L. King, "An efficient pan-sharpening method via a combined adaptive PCA approach and contourlets," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 5, pp. 1323–1335, May 2008.

[6] S. Lolli, L. Alparone, A. Garzelli, and G. Vivone, "Haze correction for contrast-based multispectral pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2255–2259, Dec. 2017.

[7] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[8] S. G. Mallat, "A theory for multiresolution signal decomposition: The wavelet representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 11, no. 7, pp. 674–693, Jul. 1989.

[9] R. A. Schowengerdt, "Reconstruction of multispatial, multispectral image data using spatial frequency content," *Photogrammetric Eng. Remote Sens.*, vol. 46, no. 10, pp. 1325–1334, Oct. 1980.

[10] M. M. Khan, J. Chanussot, L. Condat, and A. Montanvert, "Indusion: Fusion of multispectral and panchromatic images using the induction scaling technique," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 1, pp. 98–102, Jan. 2008.

[11] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2015.

[12] J. Nunez, X. Otazu, O. Fors, A. Prades, V. Pala, and R. Arbiol, "Multiresolution-based image fusion with additive wavelet decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1204–1211, May 1999.

[13] X. Fu, Z. Lin, Y. Huang, and X. Ding, "A variational pan-sharpening with local gradient constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 10257–10266.

[14] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 1753–1761.

[15] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[16] X. Zhao et al., "Progressive reconstruction network with adaptive frequency adjustment for pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17382–17397, 2024.

[17] Y.-W. Zhuo, T.-J. Zhang, J.-F. Hu, H.-X. Dou, T.-Z. Huang, and L.-J. Deng, "A deep-shallow fusion network with multidetail extractor and spectral attention for hyperspectral pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7539–7555, 2022.

[18] L. He, W. Zhang, J. Shi, and F. Li, "Cross-domain association mining based generative adversarial network for pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7770–7783, 2022.

[19] S. Li, Y. Tian, C. Wang, H. Wu, and S. Zheng, "Cross spectral and spatial scale non-local attention-based unsupervised pansharpening network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4858–4870, 2023.

[20] W. Li, X. Zhi, Y. Peng, and Y. Hu, "Pansharpening via detail guided and global scale convolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 14687–14703, 2024.

[21] B. Aiazzi, S. Baronti, M. Selva, and L. Alparone, "Bi-cubic interpolation for shift-free pan-sharpening," *ISPRS J. Photogrammetry Remote Sens.*, vol. 86, pp. 65–76, Dec. 2013.

[22] Y. Zhang, C. Liu, M. Sun, and Y. Ou, "Pan-sharpening using an efficient bidirectional pyramid network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5549–5563, Aug. 2019.

[23] M. Zhou, K. Yan, X. Fu, A. Liu, and C. Xie, "PAN-Guided band-aware multi-spectral feature enhancement for pan-sharpening," *IEEE Trans. Comput. Imag.*, vol. 9, pp. 238–249, 2023.

[24] G. Yang et al., "PanFlowNet: A flow-based deep network for pan-sharpening," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 16811–16821.

[25] L.-J. Deng, G. Vivone, C. Jin, and J. Chanussot, "Detail injection-based deep convolutional neural networks for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6995–7010, Aug. 2021.

[26] L. He et al., "Pansharpening via detail injection based convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1188–1204, Apr. 2019.

[27] H. Tao, G. Yuan, Z. Hua, and J. Li, "PDAF: Prompt-driven dynamic adaptive fusion network for pansharpening remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13533–13546, 2024.

[28] H. Lu et al., "Cross-scale interaction with spatial-spectral enhanced window attention for pansharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11521–11535, 2024.

[29] L.-J. Deng et al., "Machine learning in pansharpening: A benchmark, from shallow to deep networks," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 279–315, Sep. 2022.

[30] G. Vivone, M. Dalla Mura, A. Garzelli, and F. Pacifici, "A benchmarking protocol for pansharpening: Dataset, preprocessing, and quality assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6102–6118, 2021.

[31] B. Aiazzi, L. Alparone, S. Baronti, A. Garzelli, and M. Selva, "MTF-tailored multiscale fusion of high-resolution MS and Pan imagery," *Photogrammetric Eng. Remote Sens.*, vol. 72, pp. 591–596, May 2006.

[32] G. Vivone, R. Restaino, and J. Chanussot, "Full scale regression-based injection coefficients for panchromatic sharpening," *IEEE Trans. Image Process.*, vol. 27, no. 7, pp. 3418–3431, Jul. 2018.

[33] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "A new pansharpening algorithm based on total variation," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 318–322, Jan. 2014.

[34] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[35] F. Palsson, J. R. Sveinsson, M. O. Ulfarsson, and J. A. Benediktsson, "MTF-Based deblurring using a Wiener filter for CS and MRA pansharpening methods," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 6, pp. 2255–2269, Jun. 2016.

[36] T. Wang, F. Fang, F. Li, and G. Zhang, "High-quality Bayesian pansharpening," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 227–239, Jan. 2019.

[37] Y. Yang, H. Lu, S. Huang, W. Wan, and L. Li, "Pansharpening based on variational fractional-order geometry model and optimized injection gains," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2128–2141, 2022.

[38] P. Wang, Z. He, B. Huang, M. D. Mura, H. Leung, and J. Chanussot, "VOGTNet: Variational optimization-guided two-stage network for multispectral and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, 2024.

[39] P. Wang et al., "Low-rank tensor completion pansharpening based on haze correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5405720.

[40] A. Garzelli and F. Nencini, "Hypercomplex quality assessment of multi/hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 662–665, Oct. 2009.

[41] L. Wald, *Data Fusion: Definitions and Architectures - Fusion of Images of Different Spatial Resolutions*. Paris, France: Les Presses de l'École des Mines, 2002.

[42] D. Lei, H. Chen, L. Zhang, and W. Li, "NLRNet: An efficient nonlocal attention ResNet for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401113.

[43] Y. Liang, P. Zhang, Y. Mei, and T. Wang, "PMACNet: Parallel multiscale attention constraint network for Pan-sharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5512805.

[44] Z. Xiong, Q. Guo, M. Liu, and A. Li, "Pan-sharpening based on convolutional neural network by using the loss function with no-reference," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 897–906, 2021.